

SEQUENTIAL ESTIMATION OF THE PERCENTILES OF EXPONENTIAL AND NORMAL DISTRIBUTIONS

Malay Ghosh: Department of Statistics, University of Florida,
Little Hall, Gainesville, FL 32611

and

Nitis Mukhopadhyay: Department of Statistics, Box U-120, 196 Auditorium Road, University of Connecticut, Storrs, CT 06269

Key words: Normal mean and variance; two-parameter exponential; sequential estimator; uniform integrability; regret; second-order expansions.

Summary: The paper considers minimum risk sequential point estimation of percentiles of both the exponential (with unknown location and scale) and the normal (with unknown mean and variance) distributions. The main findings consist of asymptotic second-order expansions of the risk function associated with the corresponding sequential estimators when the loss is measured by squared error plus linear cost of sampling.

1. INTRODUCTION

The median survival time plays important roles as a summary statistic in order to understand the efficacy of treatment in various medical studies. In general, estimation of percentiles of distributions usually forms the basis in setting up appropriate tolerance limits in many reliability studies. One may refer to Brookmeyer and Crowley (1982), Gardiner et al. (1987), Bain (1978), and Englehardt and Bain (1978). The existing literature on

various aspects of estimation of the mean or the variance of a normal distribution is fairly exhaustive as can be seen from Robbins (1959), Starr and Woodroffe (1969, 1972) and Woodroffe (1977). The literature on various aspects of estimation in a two-parameter exponential population is also quite vast. One may refer to Basu (1971) and the recent review article of Mukhopadhyay (1988) in this context. Also see Govindarajulu (1987).

The present paper considers sequential point estimation of percentiles of an exponential distribution (with unknown location and scale), and a normal distribution (with unknown mean and variance). In both cases, we consider the squared error plus the linear cost of sampling as the loss function. The exponential case is discussed in Section 2, while the normal case is handled in Section 3. The emphasis is on finding the asymptotic (as the cost per unit sample gets small) second-order expansions of the risks associated with the corresponding sequential estimators of the percentiles. Our major findings have been summarized in Theorems 2.1 and 3.1, and these show that the difference between the sequential risk and the optimal fixed-sample risk is asymptotically an appropriate computable multiplier of the cost per unit sample. In other words, the "regret" (Robbins (1959)) in using the sequential estimators in the absence of any fixed sample optimal rule amounts to a specific number, which may be a fraction, of samples asymptotically.

2. PERCENTILES OF EXPONENTIAL

Let $X_1, X_2, \dots$ be independent and identically distributed (iid) random variables (rv's) with the common pdf

$$(2.1) \quad f_{\mu, \sigma}(x) = \sigma^{-1} \exp\{-(x - \mu)/\sigma\} I(x \geq \mu).$$

In the above, and in what follows, I denotes the usual indicator function. It is assumed that $\mu \in (-\infty, \infty)$ and $\sigma \in (0, \infty)$, and these are unknown parameters. The percentiles of an exponential distribution are expressible as $\mu + b\sigma$, $0 < b < \infty$. We consider the more general problem of estimating $\theta = a\mu + b\sigma$, where a $\epsilon[1, \infty)$ and $b \epsilon(0, \infty)$ are known numbers. The loss is assumed to be squared error plus cost. We write $T_n = \min\{X_1, \dots, X_n\}$

and $s_n = (n-1)^{-1} \sum_{i=1}^n (X_i - T_n)$ for $n \geq 2$. Having recorded $X_1, \dots, X_n$ we estimate θ by $\hat{\theta}_n = a(T_n - n^{-1}s_n) + bs_n$ which is unbiased for θ . Suppose that the loss incurred in estimating θ by $\hat{\theta}_n$ is given by

$$(2.2) \quad L_n = (\hat{\theta}_n - \theta)^2 + cn,$$

where $c(> 0)$ is a known constant that plays the role of cost per unit sampling. Note that the associated risk is $R_n = E(L_n) = a^2\sigma^2n^{-2} + b^2\sigma^2(n^{-1} + n^{-2}) - 2ab\sigma^2n^{-2} + cn + o(n^{-2})$ which is (approximately) minimized, had σ been known, if we take $n = n^* = (b^2\sigma^2/c)^{\frac{1}{2}}$ assuming for simplicity that n^* is an integer. Note that n^* is really obtained by minimizing $b^2\sigma^2n^{-1} + cn$. In the absence of fixed sample size rules, we propose the following stopping time. Let

$$(2.3) \quad N = \inf\{n \geq n_0 : n \geq (b^2/c)^{\frac{1}{2}} s_n\},$$

where $n_0(\geq 2)$ is the starting sampling size. Once the sampling stops, we estimate θ by $\hat{\theta}_N$. The associated risk is given by

$$\begin{aligned} (2.4) \quad R(c) &= E(L_N) \\ &= E\{a(T_N - \mu - \frac{\sigma}{N}) + (b - \frac{a}{N})(s_N - \sigma)\}^2 + cE(N) \\ &= a^2\sigma^2E(N^{-2}) + E\{b^2(s_N - \sigma)^2 - 2ab(s_N - \sigma)^2N^{-1} \\ &\quad + a^2(s_N - \sigma)^2N^{-2}\} + cE(N), \end{aligned}$$

since T_n and $I(N = n)$ are independent for every fixed $n \geq 2$. Following Lombard and Swanepoel (1978) and Swanepoel and van Wyk (1982), it can be shown that N has the same distribution as $Q + 1$ where

$$(2.5) \quad Q = \inf\{n \geq n_0 - 1 : n(n+1)/n^* \geq S_n\},$$

and $S_n = \sum_{i=1}^n W_i$ where $W_1, W_2, \dots$ are iid with the pdf $f_{0,1}(w)$.

Let $\bar{W}_Q = Q^{-1}S_Q$. Also, invoking results like Lemma 1 of Mukhopadhyay and Ekwo (1987), one can see that s_N and $\bar{W}_Q$ are identically distributed. The stopping rule (2.5) agrees with the stopping rule (2.4) of Mukhopadhyay (1988). Let

$$(2.6) \quad D = \sum_{n=1}^{\infty} n^{-1} E(S_n - 2n)^+, \nu = 1 - D, \eta = -1 - D$$

where x^+ stands for $\max(0, x)$. In what follows, we wish to examine the closeness between the sequential risk $R(c)$ and the optimal fixed sample risk R_{n^*} . In Theorem 2.1, we derive the expansion of the "regret", that is $R(c) - R_{n^*}$, up to terms of order $o(c)$. The procedure (2.3) can then be obviously seen to be asymptotically "risk efficient" in the Robbins (1959) sense.

The main result of this section is summarized as follows.

Theorem 2.1. If $0 < \sigma < \infty$, then for the stopping rule proposed in (2.3), we have as $c \rightarrow 0$:

$$(2.7) \quad R(c) = 2(b\sigma)c^{\frac{1}{2}} + c(a^2 + 4b^2 - 2ab)/b^2 + o(c)$$

if $n_0 \geq 4$, where $R(c)$ had been defined in (2.4).

First, we state and prove the following Lemmas. Throughout, we write $K(> 0)$ to denote a generic constant independent of c .

Lemma 2.1. Let $0 < \epsilon < 1$, $s(> 0)$ and $d(> 0)$ be arbitrary, but fixed numbers. Then, for the stopping rule (2.5), we have:

$$(2.8) \quad P(Q \leq \epsilon n^*) = o(c^{\frac{1}{2}(n_0-1)}) \text{ as } c \rightarrow 0;$$

$$(2.9) \quad E(Q) = n^* - 1 - D + o(1) \text{ as } c \rightarrow 0, \text{ if } n_0 \geq 3;$$

$$(2.10) \quad E\{n^{*s}(Q+d)^{-s}\} = 1 - s\{d + \eta - \frac{1}{2}(s+1)\}n^{*-1} + o(n^{*-1}) \\ \text{as } c \rightarrow 0, \text{ if } n_0 > s+4;$$

$$(2.11) \quad (n^*/Q)^s \text{ is uniformly integrable in } c(\leq c_0), \\ \text{if } n_0 > s+1;$$

$$(Q/n^*)^s \text{ is uniformly integrable in } c(\leq c_0) \\ \text{and } |Q^{-\frac{1}{2}}(S_Q - Q)|^s \text{ is uniformly integrable}$$

$$(2.12) \quad \text{in } c(\leq c_0) \text{ if } n_0 \geq 2;$$

the constants D, η being defined in (2.6).

Proof of Lemma 2.1. (2.8) and (2.10) can be proved along the lines of

Lemma 1 in Ghosh and Mukhopadhyay (1990), while (2.9) follows directly from Theorem 3 of Mukhopadhyay (1988).

In order to prove (2.11) first note that $(n^*/Q)^s I(Q > \frac{1}{2}n^*) \leq 2^s$ and thus $(n^*/Q)^s I(Q > \frac{1}{2}n^*)$ is uniformly integrable in c . On the other hand,

$$\begin{aligned} E\{(n^*/Q)^s I(Q \leq \tfrac{1}{2}n^*)\} &\leq \{n^*/(n_0 - 1)\}^s P(Q \leq \tfrac{1}{2}n^*) \\ &= O(c^{\frac{1}{2}(n_0-1-s)}), \end{aligned}$$

in view of (2.8). Thus, $\sup_{0 < c \leq c_0} E\{(n^*/Q)^s I(Q \leq \frac{1}{2}n^*)\} \leq Kc_0^{\frac{1}{2}(n_0-1-s)}$. So, if $n_0 > s + 1$, for any given $\epsilon (> 0)$, one can make $Kc_0^{\frac{1}{2}(n_0-1-s)} < \epsilon$ by suitable choice of c_0 . Hence, (2.11) follows.

Next, we turn to (2.12). Let $Z = \sup_{n \geq 2} \bar{W}_n$, and now the dominated ergodic theorem (Wiener (1939)) implies that $E(Z^s) < \infty$ for all $s > 0$. From (2.5), we get $Q < \bar{W}_{Q-1} + (n_0 - 1)$ and hence, $(Q/n^*)^s < (Z + 0(1))^s$. Thus, by Lebesgue dominated convergence theorem, $\limsup_{c \rightarrow 0} E\{(Q/n^*)^s\} \leq 1$. The uniform integrability of $(Q/n^*)^s$ follows since $Q/n^* \xrightarrow{P} 1$ as $c \rightarrow 0$. The second part of the statement in (2.12) follows now from Theorem 2 of Chow et al. (1979).

Lemma 2.2 For the stopping rule (2.3), if $n_0 \geq 4$, then we have as $c \rightarrow 0$:

$$(2.13) \quad E(N^{-2}) = n^{*-2} \{1 + o(1)\};$$

$$(2.14) \quad E\{(s_N - \sigma)^2 N^{-2}\} = o(n^{*-2});$$

$$(2.15) \quad E\{(s_N - \sigma)^2 N^{-1}\} = \sigma^2 n^{*-2} + o(n^{*-2});$$

$$(2.16) \quad E\{(s_N - \sigma)^2\} = \sigma^2 n^{*-1} + \sigma^2(4 + D)n^{*-2} + o(n^{*-2});$$

the constant D being defined in (2.6).

Proof of Lemma 2.2. We obtain (2.13) immediately from (2.11). To prove (2.14), first note that $Q^{-\frac{1}{2}}(S_Q - Q) \xrightarrow{L} N(0, 1)$ as $c \rightarrow 0$, and write

$$\begin{aligned} (2.17) \quad E\{(s_N - \sigma)^2 N^{-2}\} &= \sigma^2 E[(S_Q - Q)^2 / \{Q^2(Q + 1)^2\}] \\ &\leq \sigma^2 (n_0 - 1)^{-\delta'} E\left\{\frac{(S_Q - Q)^2}{Q} Q^{-3+\delta'}\right\}, \end{aligned}$$

for arbitrary $0 < \delta' < 1$. For $0 < \delta < 1$, Hölder's inequality and (2.17) lead to

$$E\{(s_N - \sigma)^2 N^{-2}\} \leq K E^{\delta/(1+\delta)} \left[\left\{ \frac{(S_Q - Q)^2}{Q} \right\}^{\frac{1+\delta}{\delta}} \right] E^{1/(1+\delta)} [Q^{(-3+\delta')(1+\delta)}]$$

and hence with $\delta = .001$ and $\delta' = .5$ we have $(\delta+1)/\delta = 1001$, $(-3+\delta')(1+\delta) = -2.5025$, and thus from (2.12), we claim that $E\{(S_Q - Q)^2/Q\}^{1001} < \infty$. Also, $E(Q^{-2.5025}) = 0(n^{*-2.5025}) = o(n^{*-2})$ by (2.11), if $n_0 > 1 + 2.5025$, that is if $n_0 \geq 4$. Now, (2.14) follows.

To prove (2.15), we write

$$E\{(s_N - \sigma)^2 N^{-1}\} = \sigma^2 E[(S_Q - Q)^2 / \{Q^2(Q+1)\}],$$

and it suffices to verify that

$$(2.18) \quad E \left\{ \frac{(S_Q - Q)^2}{Q} \frac{n^{*2}}{Q(Q+1)} \right\} = 1 + o(1) \text{ if } n_0 \geq 4.$$

Now,

$$\frac{(S_Q - Q)^2}{Q} \frac{n^{*2}}{Q(Q+1)} I(Q > \frac{1}{2}n^*) \leq K(S_Q - Q)^2/Q$$

and, hence, the left-hand side is uniformly integrable for $n_0 \geq 3$. See Theorem 3 of Mukhopadhyay (1988).

Thus, $E \left\{ \frac{(S_Q - Q)^2}{Q} \frac{n^{*2}}{Q(Q+1)} I(Q > \frac{1}{2}n^*) \right\} = 1 + o(1)$ for $n_0 \geq 3$. On the other hand, for $0 < \delta < 1$, Hölder's inequality lead to

$$(2.19) \quad E \left\{ \frac{(S_Q - Q)^2}{Q} \frac{n^{*2}}{Q(Q+1)} I(Q \leq \frac{1}{2}n^*) \right\} \\ \leq E^{\delta/(1+\delta)} \left[\left\{ \frac{(S_Q - Q)^2}{Q} \right\}^{(1+\delta)/\delta} \right] E^{1/(1+\delta)} \left[(n^*/Q)^{2+2\delta} I(Q \leq \frac{1}{2}n^*) \right] \\ = o(1) \{0(n^{*2+2\delta})0(c^{\frac{1}{2}(n_0-1)})\}^{1/(1+\delta)},$$

by appealing to (2.12), Theorem 3 of Mukhopadhyay (1988), and (2.8) for $n_0 \geq 2$ and $\delta = \frac{1}{4}$. Notice that the order in (2.19) with $\delta = \frac{1}{4}$ is indeed $o(1)$ if $n_0 \geq 4$, and hence we obtain (2.18).

Now, we turn to prove (2.16). Use Chow et al. (1965) and note that

$$\begin{aligned}
 E\{(s_N - \sigma)^2\} &= \sigma^2 E\{(S_Q - Q)^2/Q^2\} \\
 (2.20) \quad &= \sigma^2 E\{(S_Q - Q)^2/n^{*2}\} + \sigma^2 E\left\{\frac{(n^{*2} - Q^2)}{Q^2 n^{*2}}(S_Q - Q)^2\right\} \\
 &= \sigma^2 E(Q)/n^{*2} + \sigma^2 J/n^{*2}, \text{ say.}
 \end{aligned}$$

Now,

$$(2.21) \quad E(Q)/n^{*2} = n^{*-1} - (1 + D)n^{*-2} + o(n^{*-2}),$$

if $n_0 \geq 3$. See (2.9).

Let $R_c = (Q+1)Qn^{*-1} - S_Q$, the excess over the boundary for the stopping rule (2.5). Notice that

$$(2.22) \quad J = E[(1 - \frac{Q^2}{n^{*2}})(S_Q - Q)^2] + E\left[\left(1 - \frac{Q^2}{n^{*2}}\right)^2 \frac{n^{*2}}{Q^2}(S_Q - Q)^2\right],$$

and

$$\begin{aligned}
 1 - \frac{Q^2}{n^{*2}} &= (1 + \frac{Q}{n^*})(1 - \frac{Q+1}{n^*} + \frac{1}{n^*}) \\
 &= (1 + \frac{Q}{n^*})(-\frac{S_Q - Q}{Q} - \frac{R_c}{Q} + \frac{1}{n^*}).
 \end{aligned}$$

Thus, we obtain

$$\begin{aligned}
 (2.23) \quad &\left(1 - \frac{Q^2}{n^{*2}}\right)^2 \frac{n^{*2}}{Q^2}(S_Q - Q)^2 \\
 &= \left(1 + \frac{Q}{n^*}\right)^2 \frac{n^{*2}}{Q^2}(S_Q - Q)^2 \left\{\frac{(S_Q - Q)^2}{Q^2} + o_p(1)\right\} \\
 &= \left(1 + \frac{Q}{n^*}\right)^2 \frac{n^{*2}}{Q^2} \frac{(S_Q - Q)^4}{Q^2} + o_p(1) \\
 &= 2 \frac{(S_Q - Q)^4}{Q^2} + 2 \frac{n^*}{Q} \frac{(S_Q - Q)^4}{Q^2} + o_p(1).
 \end{aligned}$$

Now, $E\{(S_Q - Q)^4/Q^2\} = 3 + o(1)$ if $n_0 \geq 3$. Techniques similar to ones used to prove (2.18) will lead to $E\{n^*(S_Q - Q)^4/Q^3\} = 3 + o(1)$ if $n_0 \geq 3$. Thus, from (2.23), one gets

$$(2.24) \quad E \left\{ \left(1 - \frac{Q^2}{n^*} \right)^2 \frac{n^{*2}}{Q^2} (S_Q - Q)^2 \right\} = 12 + o(1) \text{ if } n_0 \geq 3.$$

Also, we have

$$(2.25) \quad \begin{aligned} \left(1 - \frac{Q^2}{n^*} \right) (S_Q - Q)^2 &= \left(1 + \frac{Q}{n^*} \right) \left(-\frac{S_Q - Q}{Q} - \frac{R_c}{Q} + \frac{1}{n^*} \right) (S_Q - Q)^2 \\ &= -(1 + \frac{Q}{n^*}) (S_Q - Q)^3 / Q - (1 + \frac{Q}{n^*}) R_c (S_Q - Q)^2 / Q \\ &\quad + (1 + \frac{Q}{n^*}) (S_Q - Q)^2 / n^*. \end{aligned}$$

We use Woodrooffe's (1977) Lemma 2.1 and techniques utilizing Hölder's inequality as before to claim that

$$(2.26) \quad \begin{cases} E\{(1 + \frac{Q}{n^*}) R_c (S_Q - Q)^2 / Q\} = 2(1 - D) + o(1) & \text{if } n_0 \geq 3, \\ E\{(1 + \frac{Q}{n^*}) (S_Q - Q)^2 / n^*\} = 2 + o(1) & \text{if } n_0 \geq 3. \end{cases}$$

Now,

$$(2.27) \quad \begin{aligned} E\{(1 + \frac{Q}{n^*}) (S_Q - Q)^3 / Q\} &= 2E[(S_Q - Q)^3 / n^*] + E \left[\frac{n^* - Q}{n^* Q} (S_Q - Q)^3 \right] \\ &= 2J_1 + J_2, \text{ say.} \end{aligned}$$

From (2.12), it follows that (for $n_0 \geq 3$)

$$(2.28) \quad \begin{aligned} J_1 &= E\{[2Q + 3Q(S_Q - Q)] / n^*\} \\ &= 2\{1 - (1 + D)n^{*-1} + o(n^{*-1})\} \\ &\quad + 3E \left[\frac{(S_Q - Q)^2}{Q} + \frac{R_c}{Q} (S_Q - Q) - \frac{S_Q - Q}{n^*} \right] \\ &= 2 + 3 + o(1) = 5 + o(1), \end{aligned}$$

by appealing to Woodrooffe's (1977) Lemma 2.1 and Hölder's inequality as before. Again, for $n_0 \geq 3$,

$$\begin{aligned}
 (2.29) \quad J_2 &= E\left[\left\{1 - \frac{S_Q + R_c}{Q} + \frac{1}{n^*}\right\}(S_Q - Q)^3/Q\right] \\
 &= E\left[-\frac{(S_Q - Q)^4}{Q^2} - \frac{R_c(S_Q - Q)^3}{Q^2} + \frac{(S_Q - Q)^3}{n^*Q}\right] \\
 &= -3 + o(1).
 \end{aligned}$$

Now, combining (2.22), (2.24) - (2.29) we obtain

$$J = 5 + 2D + o(1) \text{ if } n_0 \geq 3,$$

and hence we can derive (2.16) from (2.20).

Remark 2.1 The result (2.16) has been proved here for $n_0 \geq 3$.

Remark 2.2 If the readers' thoughts are that (2.14) and (2.15) could have been derived from (2.16), then they are possibly quite correct. However, one does not quite achieve extra elegance or save much space by sparing technicalities by doing so. For example, in order to prove (2.14), one may first write (with $Y = (s_N - \sigma)^2/n^*$)

$$E\{(s_N - \sigma)^2 N^{-2}\} = E(Y) + E\left\{Y \left(\frac{n^{*2} - N^2}{N^2}\right)\right\}.$$

Now, $E(Y)$ has been evaluated in (2.16), however, to analyze $E\left\{Y \left(\frac{n^{*2} - N^2}{N^2}\right)\right\}$, one has to go through arguments as well as details which are very similar to ours in principle. Moreover, the intricacies involved in proving (2.14) - (2.16) are quite different in each case, and hence we refrain from deriving one proof using another.

Proof of Theorem 2.1. Let $n_0 \geq 4$, then combine Lemma 2.2 with (2.4) to write

$$\begin{aligned}
 R(c) &= a^2\sigma^2(1 + o(1))/n^{*2} + b^2\sigma^2/n^* + b^2\sigma^2(4 + D)/n^{*2} \\
 &\quad - 2ab\sigma^2/n^{*2} + o(n^{*-2}) + c\{n^* - D + o(1)\} \\
 &= \{b^2\sigma^2/n^* + cn^*\} + \sigma^2\{a^2 + b^2(4 + D) - 2ab\}/n^{*2} \\
 &\quad - cD + o(c) \\
 &= 2b\sigma c^{\frac{1}{2}} + c(a^2 + 4b^2 - 2ab)/b^2 + o(c).
 \end{aligned}$$

Remark 2.3. Mukhopadhyay (1987) dealt with the same problem and obtained Theorem 2.1 in the special case when $a = b = 1$. The analysis in Theorem 2.1 is non-trivial for arbitrary a and b . In the special case when $a = b = 1$, the associated risk function can essentially be reduced to Woodrooffe's (1977) analysis, and this becomes clear if one examines equation (2.16) in Mukhopadhyay (1987). For arbitrary a and b , our treatment is both different and direct.

3. PERCENTILES OF NORMAL

Let $X_1, X_2, \dots$ be iid $N(\mu, \sigma^2)$ where both $\mu \in (-\infty, \infty)$ and $\sigma \in (0, \infty)$ are assumed to be unknown parameters. Let $\theta = a\mu + b\sigma$ where a and b are two fixed real numbers. The problem is estimation of θ under squared error loss plus cost. We write

$$\bar{X}_n = n^{-1} \sum_{i=1}^n X_i \text{ and } s_n^2 = (n-1)^{-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \text{ for } n \geq 2.$$

Having recorded $X_1, \dots, X_n$ we estimate θ by $\hat{\theta}_n = a\bar{X}_n + bs_n$. Suppose that the loss incurred in estimating θ by $\hat{\theta}_n$ is given by

$$(3.1) \quad L_n = (\hat{\theta}_n - \theta)^2 + cn$$

where $c(> 0)$ is a known constant that plays the role of cost per unit sampling. Note that the associated risk is $R_n = E(L_n) = a^2\sigma^2n^{-1} + 2b^2\sigma^2(1 - d_n) + cn$ where $d_n = \sqrt{\frac{2}{n-1}}\Gamma(n/2)/\Gamma(\frac{n-1}{2})$. Observe that $d_n = 1 - (4n)^{-1} + O(n^{-2})$, and thus

$$(3.2) \quad R_n = \sigma^2 mn^{-1} + cn + O(n^{-2})$$

where $m = a^2 + \frac{1}{2}b^2$. R_n is (approximately) minimized, had σ been known, if we take $n = n^{**} = (m\sigma^2/c)^{\frac{1}{2}}$, assuming for simplicity that n^{**} is an integer. Note that n^{**} is really obtained by minimizing $\sigma^2 mn^{-1} + cn$. Since fixed sample size rules are not available, we propose the following stopping time. Let

$$(3.3) \quad N = \inf\{n \geq n_0 : n \geq (ms_n^2/c)^{\frac{1}{2}}\},$$

where $n_0 (\geq 2)$ is the starting sample size. Once the sampling stops, we estimate θ by $\hat{\theta}_N$. The associated risk is given by

$$\begin{aligned} (3.4) \quad R^*(c) &= E(L_N) \\ &= E\{a(\bar{X}_N - \mu) + b(s_N - \sigma)\}^2 + cE(N) \\ &= a^2\sigma^2E(N^{-1}) + b^2E\{(s_N - \sigma)^2\} + cE(N), \end{aligned}$$

since $\bar{X}_n$ and $I(N = n)$ are independent for every fixed $n \geq 2$. Using Helmert's orthogonal transformation, it can be shown that N and $Q + 1$ have identical distributions where

$$(3.5) \quad Q = \inf\{n \geq n_0 - 1 : (n+1)^2 n / n^{**2} \geq S_n\},$$

and $S_n = \sum_{i=1}^n W_i$ where $W_1, W_2, \dots$ are iid χ_1^2 . In fact, $s_N^2 = \bar{W}_Q (= Q^{-1} \sum_{i=1}^Q W_i)$. The stopping rule (3.5) agrees with the stopping rule (2.4) of Mukhopadhyay (1988) with appropriate design constants. Let

$$(3.6) \quad D' = \sum_{n=1}^{\infty} n^{-1} E(S_n - 3n)^+, \nu' = \frac{3}{2} - D', \eta' = -1 - \frac{1}{2} D'$$

where recall that $x^+ = \max(0, x)$. In what follows, we now examine the closeness between the sequential risk $R^*(c)$ and the optimal fixed sample risk. In Theorem 3.1, we derive the expansion of the "regret" up to terms of order $o(c)$. From this expansion, one can easily verify that the sequential estimator $\hat{\theta}_N$ is asymptotically "risk efficient."

The main result of this section is summarized as follows.

Theorem 3.1 If $0 < \sigma < \infty$, then for the stopping rule proposed in (3.3), we have as $c \rightarrow 0$:

$$(3.7) \quad R^*(c) = 2(mc)^{\frac{1}{2}}\sigma + c \left(\frac{a^2}{2} + \frac{35b^2}{16} \right) / (a^2 + \frac{1}{2}b^2) + o(c)$$

if $n_0 \geq 6$ where $R^*(c)$ had been defined in (3.4).

First, we state and prove the following Lemmas. As before, throughout this Section, $K(> 0)$ stands for a generic constant independent of c .

Lemma 3.1. Let $0 < \epsilon < 1, s(> 0)$ and $d(> 0)$ be arbitrary, but fixed numbers. Then for the stopping rule (3.5), we have:

$$(3.8) \quad P(Q \leq \epsilon n^{**}) = O(c^{\frac{1}{2}(n_0-1)}) \text{ as } c \rightarrow 0;$$

$$(3.9) \quad E(Q) = n^{**} - 1 - \frac{1}{2}D' + o(1) \text{ as } c \rightarrow 0 \text{ if } n_0 \geq 3;$$

$$(3.10) \quad E\{n^{**s}(Q+d)^{-s}\} = 1 - s\{d + \eta' - \frac{1}{4}(s+1)\}n^{**-1} + o(n^{**-1})$$

$$\text{as } c \rightarrow 0 \text{ if } n_0 > s+4;$$

$$(3.11) \quad (n^{**}/Q)^s \text{ is uniformly integrable in } c(\leq c_0) \text{ if } n_0 > s+1;$$

$$(3.12) \quad (Q/n^{**})^s \text{ is uniformly integrable in } c(\leq c_0) \text{ and } |Q^{-\frac{1}{2}}(S_Q - Q)|^s \text{ is uniformly integrable in } c(\leq c_0) \text{ if } n_0$$

the constants D', η' being defined in (3.6).

We omit its proof. See Lemma 2.1 and its proof for some details.

Lemma 3.2. For the stopping rule (3.5), we have as $c \rightarrow 0$:

$$(3.13) \quad E\{(\bar{W}_Q - 1)^2\} = 2n^{**-1} + (3+D)n^{**-2} + o(n^{**-2}), \text{ if } n_0 \geq 3;$$

$$(3.14) \quad E\{(\bar{W}_Q - 1)^3\} = -4n^{**-2} + o(n^{**-2}), \text{ if } n_0 \geq 5;$$

$$(3.15) \quad E\{(\bar{W}_Q - 1)^4\} = 12n^{**-2} + o(n^{**-2}), \text{ if } n_0 \geq 5;$$

the constant D' being defined in (3.6).

Proof of Lemma 3.2. First note that $(2Q)^{-\frac{1}{2}}(S_Q - Q) \xrightarrow{L} N(0, 1)$ as $c \rightarrow 0$. To prove (3.13), we write

$$(3.16) \quad E\{(\bar{W}_Q - 1)^2\} = E\left\{\frac{(S_Q - Q)^2}{n^{**2}}\right\} + E\left\{\frac{n^{**2} - Q^2}{n^{**2}Q^2}(S_Q - Q)^2\right\}$$

$$= I_1 + I_2 n^{**-2}, \text{ say}$$

Now, we use (3.9) and get

$$(3.17) \quad I_1 = 2E(Q)/n^{**2} = 2n^{**-1} - (2+D')n^{**-2} + o(n^{**-2})$$

if $n_0 \geq 3$. Let $R_c = (Q+1)^2 Q n^{**-2} - S_Q$, that is the excess over the boundary for the stopping variable in (3.5). Next note that

$$(3.18) \quad \frac{n^{**2} - Q^2}{Q^2} = \left(1 - \frac{Q^2}{n^{**2}}\right) + \left(1 - \frac{Q^2}{n^{**2}}\right)^2 \frac{n^{**2}}{Q^2},$$

and

$$1 - \frac{Q^2}{n^{**2}} = -\frac{S_Q - Q}{Q} - \frac{R_c}{Q} + \frac{2Q + 1}{n^{**2}},$$

and hence we get

$$\left(1 - \frac{Q^2}{n^{**2}}\right)^2 \frac{n^{**2}}{Q^2} (S_Q - Q)^2 = \frac{n^{**2}}{Q^2} \frac{(S_Q - Q)^4}{Q^2} + o_p(1).$$

Now, we can use techniques similar to the ones utilized in proving (2.24) to claim that

$$(3.19) \quad E\left\{\left(1 - \frac{Q^2}{n^{**2}}\right)^2 \frac{n^{**2}}{Q^2} (S_Q - Q)^2\right\} = 12 + o(1) \text{ if } n_0 \geq 4.$$

On the other hand, we have

$$\left(1 - \frac{Q^2}{n^{**2}}\right) (S_Q - Q)^2 = -\frac{(S_Q - Q)^3}{Q} - R_c \frac{(S_Q - Q)^2}{Q} + \frac{(2Q + 1)}{n^{**}} \frac{(S_Q - Q)^2}{Q}$$

and we can prove that, for $n_0 \geq 3$,

$$(3.20) \quad \begin{aligned} E\left\{\left(1 - \frac{Q^2}{n^{**2}}\right)(S_Q - Q)^2\right\} &= -8 - 2\nu' + 4 + o(1) \\ &= -7 + 2D' + o(1), \end{aligned}$$

along the lines of (2.16). Now, combine (3.16), (3.17), (3.19) and (3.20) to obtain (3.13), since $I_2 = 5 + 2D' + o(1)$.

In order to prove (3.14), we write

$$(3.21) \quad \begin{aligned} E\{(\bar{W}_Q - 1)^3\} &= E\left\{\frac{(S_Q - Q)^3}{n^{**3}}\right\} + E\left\{\frac{n^{**3} - Q^3}{n^{**3}Q^3}(S_Q - Q)^3\right\} \\ &= I_3 + I_4 n^{**^{-2}}, \text{ say.} \end{aligned}$$

I_3 can be handled in the same way we have handled J_1 in (2.27) so that we have

$$(3.22) \quad I_3 = 14n^{**2} + o(n^{**2}) \text{ if } n_0 \geq 3.$$

Now,

$$(3.23)$$

$$\begin{aligned} I_4 &= E \left\{ \frac{(n^{**2} - Q^2)(n^{**2} + Qn^{**} + Q^2)}{n^{**}(n^{**} + Q)Q^3} (S_Q - Q)^3 \right\} \\ &= E \left[\frac{\{n^{**2} - (Q+1)^2 + 2Q + 1\} \{n^{**2} + Qn^{**} + Q^2\}}{n^{**}(n^{**} + Q)Q^3} (S_Q - Q)^3 \right] \\ &= E \left[-\frac{n^{**}(n^{**2} + Qn^{**} + Q^2)}{Q^2(n^{**} + Q)} \frac{(S_Q - Q)^4}{Q^2} \right. \\ &\quad \left. + \frac{2Q + 1}{n^{**}(n^{**} + Q)Q^3} (S_Q - Q)^3 \right. \\ &\quad \left. - \frac{n^{**}(n^{**2} + Qn^{**} + Q^2)}{(n^{**} + Q)Q^4} R_c(S_Q - Q)^3 \right] \\ &= E(-I_{41}) + E(I_{42}) + E(I_{43}), \text{ say.} \end{aligned}$$

Now, $I_{41}I(Q > \frac{1}{2}n^{**}) \leq K \frac{(S_Q - Q)^4}{Q^2}$ which implies that $I_{41}I(Q > \frac{1}{2}n^{**})$ is uniformly integrable if $n_0 \geq 3$. See (3.12) and Theorem 3 of Mukhopadhyay (1988). Thus, $E\{I_{41}I(Q > \frac{1}{2}n^{**})\} = \frac{3}{2}(12) + o(1) = 18 + o(1)$ if $n_0 \geq 3$. On the other hand, $E\{I_{41}I(Q \leq \frac{1}{2}n^{**})\} = o(1)$ if $n_0 \geq 5$, which can be verified along the lines of (2.19). So,

$$(3.24) \quad E(I_{41}) = 18 + o(1) \text{ if } n_0 \geq 5.$$

Similar techniques will lead to the following claims:

$$(3.25) \quad \begin{cases} E(I_{42}) = o(1) & \text{if } n_0 \geq 3, \\ E(I_{43}) = o(1) & \text{if } n_0 \geq 4. \end{cases}$$

Now, combine (3.21) - (3.25) to obtain (3.14).

In order to prove (3.15), we write

$$\begin{aligned}
 E\{(\bar{W}_Q - 1)^4\} &= \frac{1}{n^{**2}} E\left\{\frac{(S_Q - Q)^4}{Q^2}\right\} + I_5 n^{**^{-2}}, \text{ say} \\
 (3.26) \quad &= 12n^{**^{-2}} + o(n^{**^{-2}}) + I_5 n^{**^{-2}} \text{ if } n_0 \geq 3.
 \end{aligned}$$

Now, we can tackle I_5 in the same way we had handled I_2 (see (3.16)) and I_{41} (see (3.23)), and prove that $I_5 = o(1)$ if $n_0 \geq 5$.

This completes the proof of Lemma 3.2.

Proof of Theorem 3.1. Use Taylor series expansion to write

$$\begin{aligned}
 s_N &= \sigma + \frac{(s_N^2 - \sigma^2)}{2\sigma} - \frac{(s_N^2 - \sigma^2)^2}{8\sigma^3} + \frac{(s_N^2 - \sigma^2)^3}{16\sigma^5} \\
 &\quad - \frac{5}{128}(s_N^2 - \sigma^2)^4 \{\phi s_N^2 + (1 - \phi)\sigma^2\}^{-7/2}
 \end{aligned}$$

for some $\phi \in (0, 1)$ and thus we obtain

$$(3.27) \quad (s_N - \sigma)^2 = \frac{\sigma^2}{4}(\bar{W}_Q - 1)^2 - \frac{\sigma^2}{8}(\bar{W}_Q - 1)^3 + \frac{5\sigma^2}{64}(\bar{W}_Q - 1)^4 + \text{Remainder Term.}$$

Hence, for $n_0 \geq 5$, Lemma 3.2 leads to

$$\begin{aligned}
 (3.28) \quad E\{(s_N - \sigma)^2\} &= \frac{\sigma^2}{4} \left(\frac{2}{n^{**}} + \frac{3 + D'}{n^{**2}} \right) - \frac{\sigma^2}{8} \left(-\frac{4}{n^{**2}} \right) + \frac{5\sigma^2}{64} \left(\frac{12}{n^{**2}} \right) + o(n^{**^{-2}}) \\
 &= \frac{1}{2}\sigma^2 n^{**^{-1}} + 35 + 4D'\sigma^2(16n^{**2})^{-1} + o(n^{**^{-2}}) \\
 &\quad + E(\text{Remainder Term}).
 \end{aligned}$$

Now, note that

$$(3.29) \quad E(N^{-1}) = E\{(Q + 1)^{-1}\} = n^{**^{-1}} + \left(\frac{1}{2} + \frac{1}{2}D'\right)n^{**^{-2}} + o(n^{**^{-2}})$$

for $n_0 \geq 6$, in view of (3.10)

Now, observe that the remainder term in (3.27) is bounded above by $K(\bar{W}_Q - 1)^5$. Also, we can use Hölder's inequality and similar techniques utilized thus far to verify that for $0 < \delta < 1$,

$$\begin{aligned}
 (3.30) \quad E\{|\bar{W}_Q - 1|^5\} &\leq K E^{\delta/(1+\delta)} \left\{ \left| \frac{S_Q - Q}{Q^{\frac{1}{2}}} \right|^{\frac{5+5\delta}{2\sigma}} \right\} E^{1/(1+\delta)} \{Q^{-\frac{1}{2}(5+5\delta)}\} \\
 &= O(1) O(n^{**-\frac{1}{2}(5+5\delta)})
 \end{aligned}$$

if $n_0 > 1 + \frac{1}{2}(5 + 5\delta)$ in view of (3.11). Thus, the order in (3.30) can be made $o(n^{**-\frac{1}{2}})$ if $n_0 \geq 4$, by choosing $\delta = \frac{1}{4}$. Now, we combine (3.4), (3.9), (3.27) - (3.30) to complete the proof of Theorem 3.1.

ACKNOWLEDGEMENT

We acknowledge the supports through the NSF Grant Number DMS-8701814 (University of Florida) and Grant Number AFOSR 89-0225 (University of Connecticut). We also thank the referee and Professor F. Lombard for their helpful comments on an earlier version.

REFERENCES

- BAIN, L.J. (1978). *Statistical Analysis of Reliability and Life-Testing Models*. New York: Marcel Dekker.
- BASU, A.P. (1971). On a sequential rule for estimating the location parameter of an exponential distribution. *Nav. Res. Logist. Qr.*, 18, 329-337.
- BROOKMEYER, R. and CROWLEY, J. (1982). A confidence interval for the median survival time. *Biometrics*, 38, 29-41.
- CHOW, Y.S., HSIUNG, C. and LAI, T. (1979). Extended renewal theory and moment convergence in Anscombe's theorem. *Ann. Prob.*, 7, 304-318.
- CHOW, Y.S., ROBBINS, H. and TEICHER, H. (1965). Moments of randomly stopped sums. *Ann. Math. Statist.*, 36, 789-799.
- ENGLEHARDT, M. and BAIN, L.J. (1978). Tolerance limits and confidence limits on reliability for the two-parameter exponential distribution. *Technometrics*, 20, 37-39.
- GARDINER, J., SUSARLA, V. and VAN RYZIN, J. (1987). On bounded

length confidence intervals for quantities of the survival distribution. *J. Statist. Plan. Inf.*, 18, 245-253.

GHOSH, M. and MUKHOPADHYAY, N. (1990). Sequential estimation of the location parameter of an exponential distribution. *Sankhyā, Ser. A*, in press.

GOVINDARAJULU, Z. (1987). *The Sequential Statistical Analysis*. Columbus, Ohio: American Sciences Press, Inc.

LOMBARD, F. and SWANEPOEL, J.W.H. (1978). On finite and infinite confidence sequences. *S. Afr. Statist. J.*, 12, 1-24.

MUKHOPADHYAY, N. (1987). Minimum risk point estimation of the mean of a negative exponential distribution. *Sankhyā, Ser. A*, 49, 105-112.

MUKHOPADHYAY, N. (1988). Sequential estimation problems for negative exponential populations. *Sixth Statist. Review (Comm. Statist. - Theory and Methods)*, 17, 2471-2506.

MUKHOPADHYAY, N. and EKWO, M.E. (1987). A note on minimum risk point estimation of the shape parameter of a Pareto distribution. *Cal. Statist. Assoc. Bull.*, 96, 69-78.

ROBBINS, H. (1959). Sequential estimation of the mean of a normal population. *Prob. and Statist.* (U. Grenander ed.), *H. Cramér Vol.*, Almqvist and Wiksell, Uppsala, 235-245.

STARR, N. and WOODROOFE, M. (1969). Remarks on sequential point estimation. *Proc. Nat. Acad. Sci, U.S.A.*, 63, 285-288.

STARR, N. and WOODROOFE, M. (1972). Further remarks on sequential estimation: The exponential case. *Ann. Math. Statist.*, 43, 1147-1154.

SWANEPOEL, J.W.H. and VAN WYK, J.W.J. (1982). Fixed-width confidence intervals for the location parameter of an exponential distribution. *Comm. Statist., Ser. A.*, 11, 1279-1289.

WIENER, N. (1939). The ergodic theorem. *Duke Math. J.*, 5, 1-18.

WOODROOFE, M. (1977). Second-order approximations for sequential point and interval estimation. *Ann. Statist.*, 5, 984-995.